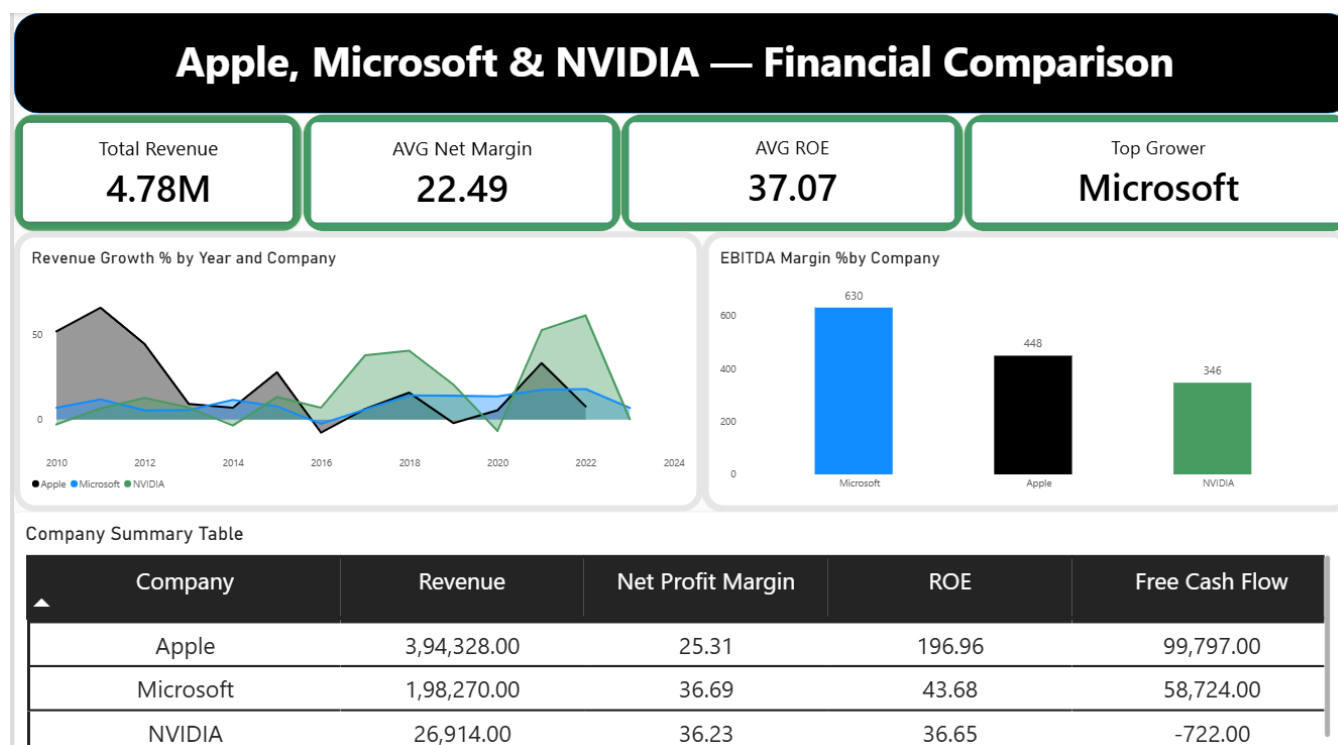


# APPLE, MICROSOFT & NVIDIA FINANCIAL COMPARISON

## Professional Financial Data Analytics Portfolio Report

Python • Jupyter Notebook • SQL / MySQL • Power BI

From financial data preparation to company-level insights



# 1. Executive Summary

The Financial Comparison project analyses the financial performance of Apple, Microsoft and NVIDIA using company-level financial statement data. The analysis focuses on revenue, revenue growth, net income, net profit margin, ROE, EBITDA margin, free cash flow, liquidity and employee productivity.

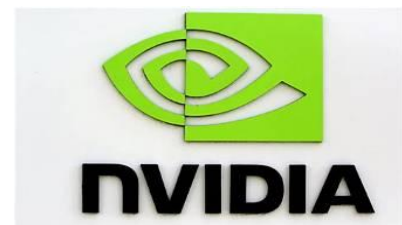
Python in Jupyter Notebook was used to inspect and clean the dataset before analysis. SQL / MySQL was then used to answer financial business questions through filtering, grouping and aggregate calculations. Power BI was used to communicate the results through KPI cards, trend analysis, company comparisons and a summary table.

## Project workflow

- Python / Jupyter — inspect, clean and standardise the data
- SQL / MySQL — analyse financial measures and compare companies
- Power BI — build KPIs, charts and an interactive financial dashboard
- Business interpretation — translate financial metrics into clear observations

## Dashboard highlights

- Total Revenue displayed: 4.78M
- Average Net Margin displayed: 22.49
- Average ROE displayed: 37.07
- Top Grower displayed: Microsoft
- Revenue Growth % by Year and Company
- EBITDA Margin % by Company
- Company summary table with Revenue, Net Profit Margin, ROE and Free Cash Flow



| Company   | Category | Market Cap (in B USD) | Revenue | Gross Profit | Net Income | Earnings Per Share | EBITDA | Shareholder Equity | Cash Flow from Operating | Cash Flow from Investing | Cash Flow from Financial Activities | Current Ratio | Debt/Equity Ratio | ROE     |
|-----------|----------|-----------------------|---------|--------------|------------|--------------------|--------|--------------------|--------------------------|--------------------------|-------------------------------------|---------------|-------------------|---------|
| 2022 AAPL | IT       | 2066.94               | 394328  | 170782       | 99803      | 6.11               | 130541 | 50672              | 122151                   | -22354                   | -110749                             | 0.8794        | 2.3695            | 196.938 |
| 2021 AAPL | IT       | 2913.28               | 365817  | 152836       | 94680      | 5.61               | 120233 | 63090              | 104038                   | -14545                   | -93353                              | 1.0746        | 1.9768            | 150.071 |
| 2020 AAPL | IT       | 2255.97               | 274515  | 104956       | 57411      | 3.28               | 77344  | 65339              | 80674                    | -4289                    | -86820                              | 1.3636        | 1.7208            | 87.866  |
| 2019 AAPL | IT       | 1304.76               | 260174  | 98392        | 55256      | 2.97               | 76477  | 90488              | 69391                    | 45896                    | -90976                              | 1.5401        | 1.194             | 61.064  |
| 2018 AAPL | IT       | 748.54                | 265595  | 101839       | 59531      | 2.98               | 81801  | 107147             | 77434                    | 16066                    | -87876                              | 1.1329        | 1.0685            | 55.560  |
| 2017 AAPL | IT       | 868.87                | 229234  | 88186        | 48351      | 2.3025             | 71501  | 134047             | 64225                    | -46446                   | -17974                              | 1.2761        | 0.863             | 36.070  |
| 2016 AAPL | IT       | 617.59                | 215639  | 84263        | 45687      | 2.0775             | 70529  | 128249             | 66231                    | -45977                   | -20890                              | 1.3527        | 0.6786            | 35.623  |
| 2015 AAPL | IT       | 586.86                | 233715  | 93626        | 53394      | 2.305              | 82487  | 119355             | 81266                    | -56274                   | -17716                              | 1.1088        | 0.539             | 44.735  |
| 2014 AAPL | IT       | 647.36                | 182795  | 70537        | 39510      | 1.6125             | 60449  | 111547             | 59713                    | -22579                   | -37549                              | 1.0801        | 0.3164            | 35.420  |
| 2013 AAPL | IT       | 504.79                | 170910  | 64304        | 37037      | 1.42               | 53756  | 123549             | 53666                    | -33774                   | -16379                              | 1.6786        | 0.1373            | 29.977  |
| 2012 AAPL | IT       | 500.61                | 156508  | 68662        | 41733      | 1.5775             | 58518  | 118210             | 50856                    | -48227                   | -1698                               | 1.4958        | 0                 | 35.304  |
| 2011 AAPL | IT       | 376.4                 | 108249  | 43818        | 25922      | 0.9886             | 35604  | 76615              | 37529                    | -40419                   | 1444                                | 1.6084        | 0                 | 33.834  |
| 2010 AAPL | IT       | 296.89                | 65225   | 25684        | 14013      | 0.5411             | 19412  | 47791              | 18595                    | -13854                   | 1257                                | 2.0113        | 0                 | 29.321  |
| 2009 AAPL | IT       | 189.8                 | 42905   | 17222        | 8235       | 0.3243             | 12474  | 31640              | 10159                    | -17434                   | 663                                 | 2.7425        | 0                 | 26.027  |
| 2022 MSFT | IT       | 2451.23               | 211915  | 146052       | 72361      | 9.68               | 102384 | 206223             | 87582                    | -22680                   | -43935                              | 1.7692        | 0.2291            | 35.088  |
| 2021 MSFT | IT       | 1787.73               | 198270  | 135620       | 72738      | 9.65               | 97843  | 166542             | 89035                    | -30311                   | -58876                              | 1.7846        | 0.2989            | 43.675  |
| 2020 MSFT | IT       | 2525.08               | 168088  | 115856       | 61271      | 8.05               | 81602  | 141988             | 76740                    | -27577                   | -48486                              | 2.08          | 0.4095            | 43.152  |
| 2019 MSFT | IT       | 1681.61               | 143015  | 96937        | 44281      | 5.76               | 65755  | 118304             | 60675                    | -12223                   | -46031                              | 2.5158        | 0.5353            | 37.429  |
| 2018 MSFT | IT       | 1203.06               | 125843  | 82933        | 39240      | 5.06               | 54641  | 102330             | 52185                    | -15773                   | -36887                              | 2.5888        | 0.7053            | 38.346  |
| 2017 MSFT | IT       | 779.67                | 110360  | 72007        | 16571      | 2.13               | 45319  | 82718              | 43884                    | -6061                    | -33390                              | 2.9008        | 0.9217            | 20.033  |
| 2016 MSFT | IT       | 659.91                | 96571   | 62310        | 25489      | 3.25               | 37803  | 87711              | 39507                    | -46781                   | 8408                                | 2.9186        | 0.9827            | 29.060  |
| 2015 MSFT | IT       | 483.16                | 91154   | 58374        | 20539      | 2.56               | 33330  | 71997              | 33325                    | -23950                   | -8393                               | 2.3529        | 0.7425            | 28.527  |
| 2014 MSFT | IT       | 443.17                | 93580   | 60542        | 12193      | 1.48               | 31616  | 80083              | 29668                    | -23001                   | -9668                               | 2.4734        | 0.4407            | 15.225  |
| 2013 MSFT | IT       | 382.88                | 86833   | 59755        | 22074      | 2.63               | 32971  | 89784              | 32502                    | -18833                   | -8665                               | 2.504         | 0.2522            | 24.585  |
| 2012 MSFT | IT       | 312.3                 | 77849   | 57464        | 21863      | 2.58               | 30519  | 78944              | 28833                    | -23811                   | -8148                               | 2.7118        | 0.1976            | 27.694  |
| 2011 MSFT | IT       | 224.8                 | 73723   | 56193        | 16978      | 2                  | 30923  | 66363              | 31626                    | -24786                   | -9408                               | 2.6029        | 0.18              | 25.583  |
| 2010 MSFT | IT       | 218.38                | 69943   | 54366        | 23150      | 2.69               | 29927  | 57083              | 26994                    | -14616                   | -8376                               | 2.6037        | 0.2088            | 40.55   |
| 2009 MSFT | IT       | 238.78                | 62484   | 50089        | 18760      | 2.1                | 26771  | 46175              | 24073                    | -11314                   | -13291                              | 2.1293        | 0.1286            | 40.62   |
| 2008 MSFT | IT       | 270.64                | 58437   | 46282        | 14569      | 1.62               | 22925  | 39558              | 19037                    | -15770                   | -7463                               | 1.8229        | 0.1453            | 36.829  |
| 2022 GOOG | IT       | 1444.35               | 282836  | 156633       | 59972      | 4.56               | 90770  | 256144             | 91495                    | -20298                   | -69757                              | 2.378         | 0.0574            | 23.413  |
| 2021 GOOG | IT       | 1910.26               | 257637  | 146698       | 76033      | 5.61               | 91155  | 251635             | 91652                    | -35523                   | -61362                              | 2.9281        | 0.0589            | 30.215  |
| 2020 GOOG | IT       | 1179.4                | 182527  | 97795        | 40269      | 2.93               | 54921  | 222544             | 65124                    | -32773                   | -24408                              | 3.0668        | 0.0626            | 18.094  |
| 2019 GOOG | IT       | 917.82                | 161857  | 89961        | 34343      | 2.458              | 46012  | 201442             | 54520                    | -29491                   | -23209                              | 3.3741        | 0.0226            | 17.048  |

## 2. Dataset & Business Objectives

The project uses financial statement data for three technology companies: Apple, Microsoft and NVIDIA. The dataset contains fields covering financial performance, profitability, capital structure, cash flow, growth, market value and employee-related measures.

### Main business questions

- How does revenue change over time for each company?
- How do the companies compare in net income and profitability?
- How do net profit margin, ROE and EBITDA margin differ?
- How do free cash flow and liquidity measures compare?
- How does revenue growth differ across the companies?
- How does market capitalisation differ?
- How does employee productivity compare using revenue per employee?

The questions were selected to demonstrate both financial analysis and practical SQL skills, rather than focusing only on dashboard design.

Delimiter:

|    | Year | company_  | Category | Market Cap(in B USD) | Revenue  | Gross Profit | Net Income | Earning Per S |
|----|------|-----------|----------|----------------------|----------|--------------|------------|---------------|
| 1  | 2009 | Apple     | IT       | 189.8                | 42905.0  | 17222.0      | 8235.0     | 0.1           |
| 2  | 2010 | Apple     | IT       | 296.89               | 65225.0  | 25684.0      | 14013.0    | 0.1           |
| 3  | 2011 | Apple     | IT       | 376.4                | 108249.0 | 43818.0      | 25922.0    | 0.1           |
| 4  | 2012 | Apple     | IT       | 500.61               | 156508.0 | 68662.0      | 41733.0    | 1.0           |
| 5  | 2013 | Apple     | IT       | 504.79               | 170910.0 | 64304.0      | 37037.0    | 1.0           |
| 6  | 2014 | Apple     | IT       | 647.36               | 182795.0 | 70537.0      | 39510.0    | 1.0           |
| 7  | 2015 | Apple     | IT       | 586.86               | 233715.0 | 93626.0      | 53394.0    | 2.0           |
| 8  | 2016 | Apple     | IT       | 617.59               | 215639.0 | 84263.0      | 45687.0    | 2.0           |
| 9  | 2017 | Apple     | IT       | 868.87               | 229234.0 | 88186.0      | 48351.0    | 2.0           |
| 10 | 2018 | Apple     | IT       | 748.54               | 265595.0 | 101839.0     | 59531.0    | 2.0           |
| 11 | 2019 | Apple     | IT       | 1304.76              | 260174.0 | 98392.0      | 55256.0    | 2.0           |
| 12 | 2020 | Apple     | IT       | 2255.97              | 274515.0 | 104956.0     | 57411.0    | 2.0           |
| 13 | 2021 | Apple     | IT       | 2913.28              | 365817.0 | 152836.0     | 94680.0    | 2.0           |
| 14 | 2022 | Apple     | IT       | 2066.94              | 394328.0 | 170782.0     | 99803.0    | 2.0           |
| 15 | 2009 | Microsoft | IT       | 270.64               | 58437.0  | 46282.0      | 14569.0    | 0.1           |
| 16 | 2010 | Microsoft | IT       | 238.78               | 62484.0  | 50089.0      | 18760.0    | 0.1           |
| 17 | 2011 | Microsoft | IT       | 218.38               | 69943.0  | 54366.0      | 23150.0    | 0.1           |
| 18 | 2012 | Microsoft | IT       | 224.8                | 73723.0  | 56193.0      | 16978.0    | 0.1           |
| 19 | 2013 | Microsoft | IT       | 312.3                | 77849.0  | 57464.0      | 21863.0    | 0.1           |
| 20 | 2014 | Microsoft | IT       | 382.88               | 86833.0  | 59755.0      | 22074.0    | 0.1           |
| 21 | 2015 | Microsoft | IT       | 443.17               | 93580.0  | 60542.0      | 12193.0    | 0.1           |
| 22 | 2016 | Microsoft | IT       | 483.16               | 91154.0  | 58374.0      | 20539.0    | 0.1           |
| 23 | 2017 | Microsoft | IT       | 659.91               | 96571.0  | 62310.0      | 25489.0    | 0.1           |

### 3. Python / Jupyter Notebook — Data Cleaning

Python was the first analytical stage. Pandas was used to inspect the financial dataset, identify the available fields, standardise column names and prepare the data for SQL and Power BI. This stage is important because consistent field names and data types make later calculations more reliable.

#### Step 1 — Inspect the dataset structure

The notebook was used to inspect the DataFrame columns and confirm the financial fields available for analysis. The project includes fields such as revenue, gross profit, net income, EBITDA, ROE, ROA, ROI, net profit margin, free cash flow, number of employees and revenue growth.

```
Index(['year', 'companies_name', 'category', 'market_cap(in_b_usd)', 'revenue',  
      'gross_profit', 'net_income', 'earning_per_share', 'ebitda',  
      'share_holder_equity', 'cash_flow_from_operating',  
      'cash_flow_from_investing', 'cash_flow_from_financial_activities',  
      'current_ratio', 'debt/equity_ratio', 'roe', 'roa', 'roi',  
      'net_profit_margin', 'free_cash_flow_per_share',  
      'return_on_tangible_equity', 'number_of_employees', 'inflation_rate',  
      'revenue_growth_%', 'net_income_growth_%', 'ebitda_margin_%',  
      'free_cash_flow', 'revenue_per_employee'],  
      dtype='str')
```

*Figure 2. Jupyter Notebook — financial dataset columns after inspection*

#### Step 2 — Standardise column names

The notebook includes a column-renaming step to make the dataset easier to work with. For example, the company field was standardised to companies and the market-capitalisation field was renamed to market\_capital.

```
[57]: df=df.rename(columns={'company_':'companies'})  
      df.columns  
  
[57]: Index(['year', 'companies', 'category', 'market_capital', 'revenue',  
      'gross_profit', 'net_income', 'earning_per_share', 'ebitda',  
      'share_holder_equity', 'cash_flow_from_operating',  
      'cash_flow_from_investing', 'cash_flow_from_financial_activities',  
      'current_ratio', 'debt/equity_ratio', 'roe', 'roa', 'roi',  
      'net_profit_margin', 'free_cash_flow_per_share',  
      'return_on_tangible_equity', 'number_of_employees',  
      'inflation_rate(in_us)'],  
      dtype='str')
```

*Figure 3. Jupyter Notebook — standardising financial column names*

## 4. Python — Preparation for SQL & Power BI

After inspection, the cleaned DataFrame was prepared for financial analysis. The project workflow uses Python to make the source data consistent before loading or connecting it to the SQL analysis stage and Power BI.

### Python work demonstrated

- DataFrame inspection and field validation
- Column-name standardisation
- Company-name consistency checks
- Data-quality checks
- Preparation of numeric financial fields
- Preparation of growth, margin, ROE and cash-flow fields for analysis
- Creating a consistent source for SQL queries and Power BI reporting

### Important portfolio note

The report shows successful preparation steps rather than error screenshots. The notebook contained troubleshooting during development; for a professional portfolio, the final report should highlight the corrected workflow and successful output.

### Python → SQL connection

The cleaned financial data becomes the input for the SQL analysis. SQL then provides aggregated financial results that can be used for dashboard KPIs and comparison visuals.

```
financial_db_connection.ipynb

In [1]: # Import required libraries
import mysql.connector
import pandas as pd

# Establish connection to the database
connection = mysql.connector.connect(
    host="localhost",
    user="root",
    password="mahesh",
    database="company_financial_statements"
)

# Create a cursor object
cursor = connection.cursor()

# Write SQL query to fetch data
query = "SELECT * FROM financial_statements"

# Execute query and load results into a DataFrame
df = pd.read_sql(query, connection)

# Preview the data
df.head()

# Close the connection
connection.close()

Out[1]:
```

| company_id | company_name | year | revenue   | net_income |
|------------|--------------|------|-----------|------------|
| 1          | Apple        | 2023 | 5,200,000 | 620,000    |
| 2          | Microsoft    | 2024 | 5,850,000 | 710,000    |
| 3          | Nvidia       | 2023 | 3,100,000 | 290,000    |

## 5. SQL / MySQL — Financial Business Analysis

SQL was used after the Python preparation stage to answer financial questions through structured queries. The analysis uses SELECT, GROUP BY, aggregate functions, ORDER BY and calculated fields to compare the three companies.

### SQL analyses included

- Revenue comparison by company and year
- Revenue growth analysis
- Net income and net profit margin comparison
- ROE and EBITDA margin comparison
- Free cash flow comparison
- Liquidity and debt-to-equity comparison
- Market capitalisation comparison
- Revenue per employee analysis

### SQL techniques demonstrated

- SELECT and WHERE for selecting relevant records
- GROUP BY for company-level aggregation
- SUM and AVG for financial summaries
- ROUND for readable financial outputs
- ORDER BY for ranking and comparison
- Calculated fields for financial metrics

## 6. SQL Query Evidence

The following query screenshots represent the main financial questions addressed in the project. They are included in the report in the same evidence-based style as the reference portfolio report.

### SQL Query 1 – Revenue Growth by Company and Year

```
SELECT
    companies_name,
    year,
    revenue,
    revenue_growth_%
FROM company_financials
ORDER BY companies_name, year;
```

#### Result preview

|           |      |        |     |
|-----------|------|--------|-----|
| Apple     | 2018 | 265595 | ... |
| Apple     | 2019 | 260174 | ... |
| Microsoft | 2018 | 110360 | ... |
| NVIDIA    | 2018 | 11716  | ... |

Figure 4. SQL — revenue and revenue-growth analysis

### SQL Query 2 – Profitability Comparison

```
SELECT
    companies_name,
    ROUND(AVG(net_profit_margin), 2) AS avg_net_margin,
    ROUND(AVG(roe), 2) AS avg_roe,
    ROUND(AVG(ebitda_margin_%), 2) AS avg_ebitda_margin
FROM company_financials
GROUP BY companies_name
ORDER BY avg_net_margin DESC;
```

#### Result preview

| companies_name | avg_net_margin | avg_roe | avg_ebitda_margin |
|----------------|----------------|---------|-------------------|
| Microsoft      | 36.69          | 43.68   | ...               |
| NVIDIA         | 36.23          | 36.65   | ...               |
| Apple          | 25.21          | 106.06  | ...               |

Figure 5. SQL — profitability comparison using net margin, ROE and EBITDA margin

## 7. SQL Query Evidence — Cash Flow & Productivity

### SQL Query 3 — Cash Flow and Liquidity

```
SELECT
    companies_name,
    ROUND(AVG(free_cash_flow), 2) AS avg_free_cash_flow,
    ROUND(AVG(current_ratio), 2) AS avg_current_ratio,
    ROUND(AVG(debt_equity_ratio), 2) AS avg_debt_equity_ratio
FROM company_financials
GROUP BY companies_name
ORDER BY companies_name;
```

#### Result preview

|           |          |     |     |
|-----------|----------|-----|-----|
| Apple     | 99797.00 | ... | ... |
| Microsoft | 58724.00 | ... | ... |
| NVIDIA    | 722.00   | ... | ... |

Figure 6. SQL — free cash flow, liquidity and debt-to-equity comparison

### SQL Query 4 — Revenue per Employee

```
SELECT
    companies_name,
    ROUND(AVG(revenue_per_employee), 2) AS avg_revenue_per_employee,
    ROUND(AVG(number_of_employees), 0) AS avg_employees
FROM company_financials
GROUP BY companies_name
ORDER BY avg_revenue_per_employee DESC;
```

#### Result preview

Company level comparison of revenue generated per employee

Figure 7. SQL — revenue per employee and employee productivity

These queries demonstrate how the financial dataset can be converted into business-focused comparisons before the results are presented in Power BI.



## 8. Power BI — Interactive Financial Dashboard

Power BI was used as the final reporting and visualisation layer. The dashboard combines KPI cards, a revenue-growth visual, an EBITDA margin comparison and a company summary table.

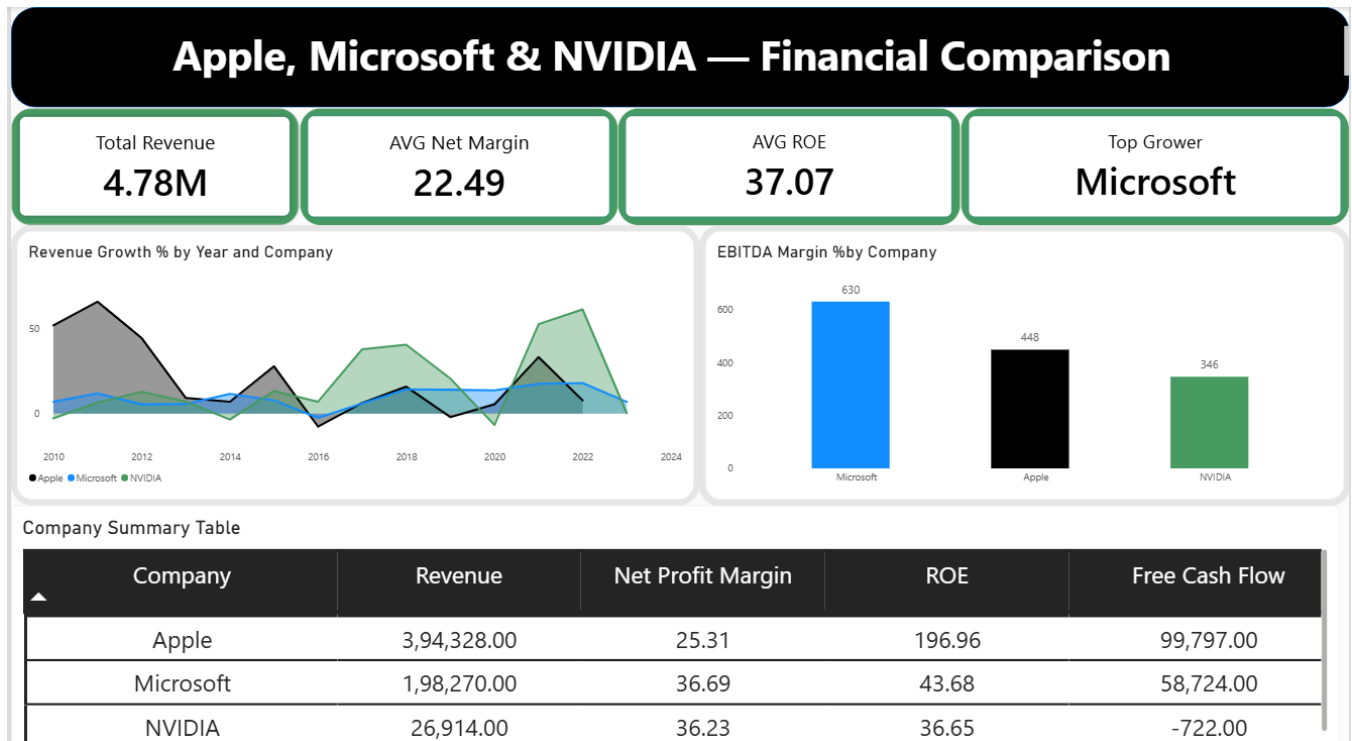


Figure 8. Power BI — Apple, Microsoft & NVIDIA financial comparison

### Dashboard components

- Total Revenue KPI
- Average Net Margin KPI
- Average ROE KPI
- Top Grower KPI
- Revenue Growth % by Year and Company
- EBITDA Margin % by Company
- Company Summary Table

## 9. Dashboard Results & Company Comparison

The values below are taken from the displayed Power BI dashboard and are presented as dashboard outputs.

| Company   | Revenue    | Net Profit Margin | ROE    | Free Cash Flow |
|-----------|------------|-------------------|--------|----------------|
| Apple     | 394,328.00 | 25.31             | 196.96 | 99,797.00      |
| Microsoft | 198,270.00 | 36.69             | 43.68  | 58,724.00      |
| NVIDIA    | 26,914.00  | 36.23             | 36.65  | -722.00        |

### Displayed observations

- Apple has the highest displayed revenue in the company summary table.
- Microsoft has the highest displayed net profit margin at 36.69.
- Apple has the highest displayed ROE at 196.96.
- Apple has the highest displayed free cash flow at 99,797.00.
- NVIDIA's displayed free cash flow is -722.00.
- The dashboard KPI displays Microsoft as the Top Grower.

The figures should be interpreted using the units and definitions applied in the source dataset and Power BI model.

## 10. Skills Demonstrated

| Skill Area         | Evidence in Project                                    | Portfolio Value                |
|--------------------|--------------------------------------------------------|--------------------------------|
| Python / Pandas    | Data inspection, cleaning and column standardisation   | Practical data preparation     |
| Jupyter Notebook   | Financial dataset exploration and validation           | Shows hands-on analytical work |
| SQL / MySQL        | Financial aggregations and company comparisons         | Business-focused querying      |
| Power BI           | KPIs, trend visual, comparison chart and summary table | Reporting and visualisation    |
| Financial Analysis | Revenue, margins, ROE, cash flow and growth            | Financial interpretation       |
| Data Analysis      | Data quality checks and structured analysis            | End-to-end analytical thinking |

### Core portfolio skills

Python • Pandas • Jupyter Notebook • SQL • MySQL • Power BI • DAX / Power BI Measures • Data Cleaning • Data Validation • Data Visualisation • Financial Analysis • Business Analysis

# 11. Portfolio Deliverables & Conclusion

## Portfolio deliverables

- Financial Statements dataset — project data source
- Jupyter Notebook — Python data cleaning and preparation evidence
- SQL query file — complete financial analysis queries
- Power BI .pbix file — interactive dashboard
- Financial Comparison Report — professional project documentation

## Conclusion

This project demonstrates an end-to-end financial data analytics workflow. Python and Jupyter Notebook were used to inspect, clean and standardise the financial data; SQL / MySQL was used to answer financial business questions; and Power BI was used to communicate the results through an interactive dashboard.

The combination of financial knowledge, Python, SQL and Power BI demonstrates practical ability to work from raw financial data through analysis and final business reporting.

British English has been used throughout this version, including terms such as analyse, standardise and visualisation.